PRODUCT REQUIREMENTS DOCUMENT (PRD)

Centralized School ERP Management System — Ever Grow Schools

Target Audience: Single Campus (1,000–3,000 Students)	**Document Version:** 1.1.0 (Production Release + Extensions)
Status: Phases 1–10 Implemented & Verified; Phases 11–12 Planned	**Stack:** FastAPI + PostgreSQL + React 19 + Electron
Document Owner: Project Development Team	**Last Updated:** October 2026

About this document

This Product Requirements Document defines the purpose, users, functional scope, architecture, security model, data design and delivery status of the Ever Grow Schools ERP platform. Version 1.1.0 keeps the verified production baseline (Phases 1–10) and extends it with two new modules: **GPS Bus Tracking & Transport Management** and an **AI Career Recommendation Engine** for Class 12 students, based on academic marks and personal interests.

11 Functional modules (Module 0 – 10)	**3** Portals: Staff, Parent and Student	**9** Role-based access personas	**12** Delivery phases (10 verified, 2 planned)

What is new in version 1.1.0

- **Module 9 — GPS Bus Tracking:** live vehicle location, route and stop management, trip lifecycle, ETA, geofence alerts and parent tracking.
- **Module 10 — AI Career Recommendation:** interest assessment plus academic marks produce ranked career fields, courses and entrance-exam guidance after Class 12.
- **Expanded documentation:** architecture, RBAC matrix, database design, API design, testing, deployment, risks, roadmap and glossary.
- **Cleaner layout:** duplicate welcome screens removed and image placeholders added so final screenshots can be inserted. Every main section now starts at the top of a new page.

Table of Contents

1. Executive Summary & Objectives . . . 3
2. Scope, Stakeholders & Assumptions . . . 4
3. User Personas & Role-Based Access Control (RBAC) . . . 5
4. System Architecture & Technology Stack . . . 7
5. Module 0: Web Application Architecture & Unified Portal Login . . . 9
6. Module 1: Executive Dashboard & Multi-Tier Analytics . . . 11
7. Module 2: Student Lifecycle, Admissions & KYC Directory . . . 12
8. Module 3: Classroom Attendance Register & Conflict-Free Timetable . . . 14
9. Module 4: Examinations, Grading Pipeline & PDF Report Cards . . . 16
10. Module 5: Fee Architecture, Concessions & Sequential Receipts . . . 18
11. Module 6: Academic Structure & Staff Directory . . . 20
12. Module 7: Segregated Parent & Student Portals . . . 21
13. Module 8: Reports & Excel Export Center . . . 22
14. Module 9: GPS Bus Tracking & Transport Management (NEW) . . . 23
15. Module 10: AI Career Recommendation for Class 12 Students (NEW) . . . 27
16. Database Design . . . 32
17. API Design . . . 34
18. Non-Functional Requirements & Security . . . 35
19. Testing & Quality Assurance . . . 36
20. Deployment & Infrastructure . . . 37
21. Implementation & Delivery Verification Status . . . 38
22. Risks & Mitigations . . . 39
23. Future Enhancements . . . 40
24. Glossary . . . 41
Appendix A: Figure and Screenshot Checklist . . . 42
Appendix B: Module Traceability Matrix . . . 43
Appendix C: Document Approval . . . 44

Revision History

Version	Date	Phases covered	Summary of changes
1.0.0	Baseline	1–10	Production release: authentication, dashboard, admissions, attendance, timetable, exams, fees, portals, exports and Electron packaging.
1.1.0	October 2026	1–12	Added Module 9 (GPS Bus Tracking) and Module 10 (AI Career Recommendation); added architecture, RBAC matrix, database, API, testing, deployment, risk and roadmap sections; removed duplicate screens and added image placeholders. Layout revised so every main section starts at the top of a page and the table of contents page numbers match.

1. Executive Summary & Objectives

The School ERP Management System is a unified, secure Enterprise Resource Planning platform developed to automate and digitize academic administration, attendance registers, conflict-free scheduling, examination grading, sequential fee collection, and parent-student engagement. It is designed for a single campus with 1,000–3,000 students and is delivered as a web application with an Electron desktop wrapper for offline campus PCs.

This version extends the verified production baseline with two forward-looking capabilities: live GPS tracking of school buses so parents and the school always know where each vehicle is, and an AI-assisted career recommendation engine that helps Class 12 students choose a suitable field of study using their marks and personal interests.

1.1 Core Objectives

- **Consolidated Single Source of Truth:** replaces disjointed spreadsheets with a normalized PostgreSQL database.
- **Sequential Financial Integrity:** enforces unmodifiable sequential receipts (RCP-YYYY-XXXXX) with immutable audit trails.
- **Automated Official Documents:** generates tamper-resistant ReportLab PDF report cards and receipts with QR verification codes.
- **Zero-Leakage Portal Security:** strictly prevents parents from accessing records outside their verified linked children.
- **Live Transport Visibility:** tracks school buses in real time, shares trip status and ETA with authorised parents and raises safety alerts.
- **Guided Career Decisions:** turns marks and interests into explainable, counsellor-approved career guidance for Class 12 students.
- **Role-Segregated Experience:** gives every persona only the screens, data and actions required for their work.

1.2 Problem Statement

Schools of this size typically run on paper registers, spreadsheets and phone calls. Attendance is recorded late, fee receipts are numbered manually, report cards are typed one by one, timetable clashes are found only when a class is left without a teacher, and parents depend on phone calls to learn about their child’s attendance, dues or bus position. Students in Class 12 receive career advice that depends on the time a counsellor can spare, and rarely on a structured analysis of their own results and interests. This platform removes these gaps with a single, auditable system.

1.3 Success Metrics

Metric	Target	How it is measured
Classroom attendance entry time	Under 2 minutes per section	Timestamps between roll-call open and submit
Duplicate or out-of-sequence receipt numbers	Zero	Database constraint checks and receipt audit report
Cross-child data leakage in parent portal	Zero incidents	Automated RBAC test suite and manual penetration checks
API response time for indexed queries	Under 200 ms	Request timing logs under test load
Report card generation	Under 5 seconds per student	PDF generation timing in the test suite
Bus location freshness during a live trip	Average under 15 seconds	Server receive time minus device timestamp
Class 12 students completing career assessment	At least 90% of the batch	Assessment completion report per academic year
Career report approved by counsellor before release	100%	Review status in the recommendation audit log

2. Scope, Stakeholders & Assumptions

2.1 In Scope and Out of Scope

In scope (this release)	Not part of this release
Web application with Staff, Parent and Student portals sharing one backend and database.	Native Android/iOS mobile applications (the driver and parent views run in the responsive web app).
Electron desktop package for offline campus PCs.	Online payment-gateway integration; fees are recorded at the cashier desk.
Modules 0–10: login, dashboard, admissions, attendance, timetable, exams, fees, academic structure, portals, exports, GPS bus tracking, AI career guidance.	SMS and WhatsApp notification automation.
PDF generation (report cards, receipts, career reports) and Excel exports.	Payroll, library and inventory management.
Single campus, 1,000–3,000 students, English interface.	Multi-branch management and Tamil-language interface (see Future Enhancements).

2.2 Stakeholders

Stakeholder	Interest in the system	Involvement
Management / Principal	Oversight of attendance, exam approvals, staff and finance.	Approver and primary report consumer.
Office & admission staff	Fast enquiry handling, clean student records and KYC.	Daily operator.
Teachers	Quick roll-call, marks entry and timetable visibility.	Daily operator.
Cashier / Accounts	Accurate fee collection, concessions and reconciliation.	Daily operator.
Parents and Guardians	Child’s attendance, dues, report cards and bus location.	Self-service user.
Students	Timetable, homework notices, grades and career guidance.	Self-service user.
Transport coordinator and drivers	Route planning, trip management and safe operation.	Operator of Module 9.
Career counsellor	Review and approve AI-generated career reports.	Reviewer in Module 10.
Development team	Maintainable code, repeatable deployment, test coverage.	Builder and maintainer.

2.3 Assumptions and Constraints

- The school provides accurate master data (classes, sections, subjects, fee structures, staff list) before go-live.
- Every student has at least one verified guardian phone number linked to the record.
- School buses are fitted with a GPS device that exposes an API, or drivers carry a smartphone with mobile data (see Module 9).
- Class 11–12 marks are entered through the Examinations module, so career guidance always uses approved and published marks.
- Personal data of minors is processed only for school purposes, with access restricted by role and recorded in audit logs.
- The system must remain usable on campus PCs with limited internet through the Electron desktop deployment.

3. User Personas & Role-Based Access Control (RBAC)

Access is organised around nine personas. Every API route verifies the caller's role on the server, so hiding a button in the interface is never the only protection. The first six roles form the verified baseline; Transport Coordinator, Driver and Career Counsellor are introduced with Modules 9 and 10.

Role	Portal	Primary Scope & Responsibilities
Super Admin	Staff Portal	Global configuration, database seeding, audit log inspection, user roles
Principal	Staff Portal	Executive oversight, attendance trends, exam approvals, staff management
Teacher	Staff Portal	Daily classroom roll-call, mark submission, timetable schedule view
Cashier	Staff Portal	Fee collection, concession application, sequential receipts, reconciliation
Parent	Parent Portal	Strict child-linked view of attendance, fee status, and published report cards
Student	Student Portal	Personal timetable matrix, homework notices, published term grades
Transport Coordinator (new)	Staff Portal	Vehicles, drivers, routes, stops, student-route mapping, live fleet view and alerts
Driver (new)	Driver view	Start and end assigned trips, share live location, mark stops as completed
Career Counsellor (new)	Staff Portal	Review AI career reports, add counsellor notes, approve release to students and parents

3.1 Access Control Principles

- **Least privilege:** each role receives only the permissions its duties require.
- **Server-side enforcement:** JWT claims carry the role; dependencies on every route reject unauthorised calls with a generic response.
- **Linked-record rule:** parents and students resolve every query through verified guardian and student links.
- **Publication gate:** marks, report cards and career reports stay invisible to families until approved and published.
- **Auditability:** sensitive actions such as attendance corrections, concessions, receipt cancellations and approvals are written to the audit log.

3.2 Permission Matrix

F = full control, W = create/edit, R = read only, O = own or linked records only, – = no access.

Module	SA	PR	TE	CA	PA	ST	TC	DR	CC
Dashboard & analytics	F	F	R	R	O	O	R	–	R
Admissions & students	F	W	R	R	O	O	R	–	R
Staff directory	F	W	O	O	–	–	O	O	O
Attendance register	F	W	W	–	O	–	–	–	–
Timetable	F	W	R	–	–	O	–	–	–
Exams & marks	F	W	W	–	O	O	–	–	R
Fees & receipts	F	R	–	W	O	–	–	–	–
Reports & exports	F	F	R	R	–	–	R	–	R
Transport & GPS	F	R	–	–	O	O	F	W	–
Career guidance	F	R	–	–	O	O	–	–	F
Audit logs	F	R	–	–	–	–	–	–	–

SA Super Admin, PR Principal, TE Teacher, CA Cashier, PA Parent, ST Student, TC Transport Coordinator, DR Driver, CC Career Counsellor.

3.3 A Day in the Life

- **Teacher:** opens the staff portal, sees today’s timetable, marks roll-call for each section in one click per student, and later submits marks for review.
- **Cashier:** searches a student, sees dues after sibling and scholarship concessions, takes cash, card or UPI, and prints the next sequential receipt.
- **Parent:** logs in with the registered phone number, checks attendance, fee status and report cards of each linked child, and follows the bus on a map during pick-up.
- **Student (Class 12):** views the timetable, reads homework notices, completes the interest assessment and later reads the counsellor-approved career report.

4. System Architecture & Technology Stack

The platform follows a layered client-server architecture. A React single-page application talks to an asynchronous FastAPI backend that applies business rules and persists to PostgreSQL. The same web build is packaged inside Electron for offline campus PCs, and the whole backend runs with Docker Compose for web and cloud deployment.

4.1 Technology Stack

Layer	Technology	Responsibility
Frontend	React 19, TypeScript, Tailwind CSS	Role-based portals, glassmorphism interface, charts and forms
Desktop	Electron with native installer	Offline-capable packaging for campus PCs
Backend API	FastAPI on async ASGI	REST endpoints, validation, authorisation, business rules
Database	PostgreSQL	Normalized relational storage with strict constraints and foreign keys
Authentication	bcrypt, JWT access and refresh tokens	Password hashing, token rotation, account lockout
Documents	ReportLab, OpenPyXL	PDF report cards, receipts and career reports; Excel exports
Maps (Module 9)	OpenStreetMap with Leaflet (provider configurable)	Live bus map, routes, stops and geofences
Real-time (Module 9)	WebSocket or Server-Sent Events	Push live vehicle positions to open dashboards
AI (Module 10)	Rule and scoring engine plus configurable LLM provider	Match scoring, explanation text and career report content
Deployment	Docker Compose	Repeatable web and cloud deployment

4.2 Layered Design

- **Presentation layer:** React pages, role-aware navigation and reusable components; no business rule lives only in the browser.
- **API layer:** FastAPI routers grouped by module, request and response schemas, dependency-based role checks.
- **Service layer:** domain logic such as conflict detection, grade calculation, receipt sequencing, ETA calculation and career scoring.
- **Data layer:** ORM models with eager loading to avoid N+1 queries, UUID v4 keys and UTC timestamp mixins.
- **Integration layer:** GPS device or driver-app ingestion, map tiles and the optional LLM provider, each behind a replaceable adapter.

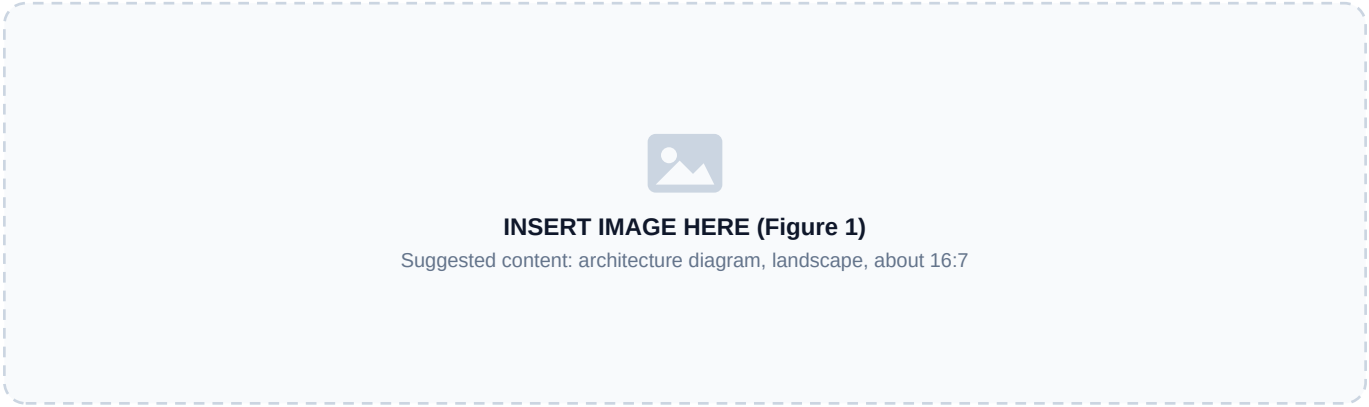


Figure 1: High-level system architecture: clients, API, services, database and integrations.

4.3 Typical Request Lifecycle

1. The user signs in through the unified login; the backend validates the identifier, applies lockout rules and issues an access and refresh token pair.
2. The browser sends every request with the access token; a dependency decodes it and checks the role against the route.
3. The service layer validates input, applies business rules inside a database transaction and writes audit entries for sensitive actions.
4. The response is serialised through typed schemas so that only permitted fields are returned.
5. Dashboards and live maps refresh through polling or the real-time channel without exposing other users’ data.

4.4 Engineering Principles

- **Immutable money records:** all financial values use Decimal(12, 2); receipts are never edited, only cancelled with a new immutable record.
- **Identifiers:** UUID v4 primary keys prevent guessable record numbers; human-friendly codes (ENQ, ADM, RCP) are separate fields.
- **Time:** all timestamps are stored in UTC and converted for display.
- **Replaceable integrations:** GPS provider, map tiles and LLM sit behind adapters so a vendor change does not touch business logic.

5. Module 0: Web Application Architecture & Unified Portal Login

PHASE	PRIMARY ROLES	STATUS
Phase 2 and Phase 9	**All roles**	**Implemented & Verified**

The web application features an Apple-inspired glassmorphism interface with segmented role tabs, dynamic login identifiers, and brand identity. A single entry point serves Staff, Parents and Students while the backend decides what each account may do.

- **Multi-Portal Segmented Tabs:** one unified login screen for Staff, Parents, and Students with automated backend role enforcement.
- **Flexible Identifier Authentication:** students log in via Registration Numbers, parents via phone number, and teachers via phone number.
- **Security Safeguards:** OWASP-compliant 5-attempt account lockout, JWT token rotation, and zero-leakage generic authentication responses.
- **Branded Experience:** integrates the official Ever Grow Schools colorful brand emblem and a short animated welcome transition with a Skip intro option.

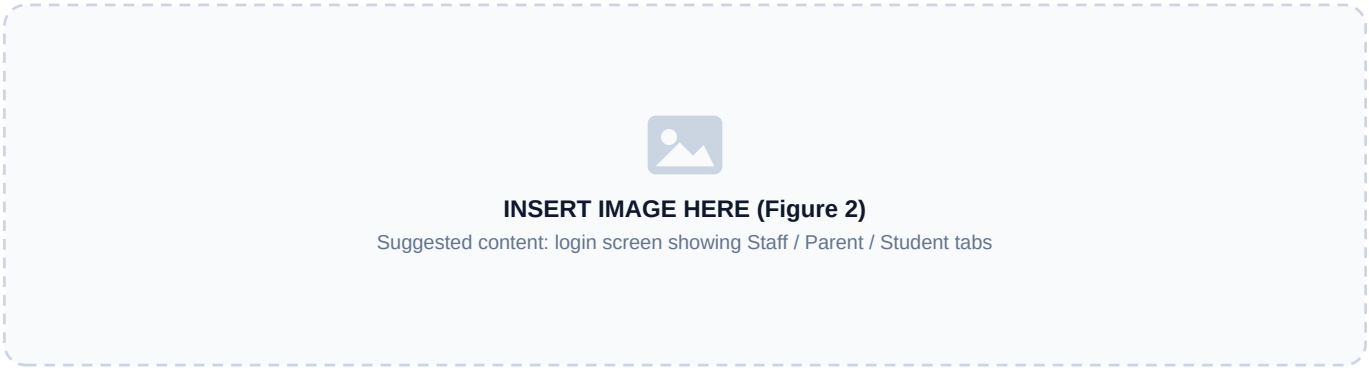


Figure 2: Unified login portal with role pill tabs and dynamic identifier input.

5.1 Identifier Rules (Seeded Test Data)

Portal tab	Identifier	Seeded test range
Student	Registration Number	1 to 10
Parent	Registered phone number	1111111111 to 5555555555
Teacher (Staff tab)	Registered phone number	6666666666 to 9999999999
Other staff (Super Admin, Principal, Cashier)	Configured staff credentials	Created by Super Admin

5.2 Authentication Flow

1. The user selects the portal tab; the form changes its identifier label and input rules (registration number or phone).
2. The server finds the account and compares the bcrypt hash; every failure returns the same generic message so that accounts cannot be guessed.
3. After five consecutive failures the account is locked and the event is recorded for the Super Admin.
4. On success the server returns a short-lived access token and a refresh token; the refresh token is rotated on every use.
5. The role in the token decides which portal routes and menu entries are shown; the same role is checked again on each API call.

5.3 Security Controls

Control	Behaviour
Password storage	bcrypt hashing; plain passwords are never stored or logged.
Account lockout	Locks after 5 failed attempts to slow brute-force attacks.
Token handling	JWT access and refresh tokens with rotation and expiry.
Error messages	Generic responses so the system never reveals whether an identifier exists.
Role enforcement	Portal tab is only a convenience; the backend rejects a role that does not match the account.

6. Module 1: Executive Dashboard & Multi-Tier Analytics

PHASE	PRIMARY ROLES	STATUS
Phase 8	**Super Admin, Principal, Cashier, Teacher**	**Implemented & Verified**

Provides real-time institutional telemetry, high-level metrics, and live operational trends for school administrators and principals. The dashboard is the landing page of the Staff Portal and links directly to the most common daily tasks.

- **Real-time KPI Widgets:** total student enrollment, daily presence rate, YTD revenue, and active staff counts.
- **Attendance Trends:** 12-month historical interactive trendlines for seasonal absence monitoring.
- **Fee Collection Donut:** visual ratio of Completed, Pending, and Overdue tuition invoices.
- **Live Application Feed:** recent admission enquiries with rapid status toggling (Enrolled, Pending, Interview).

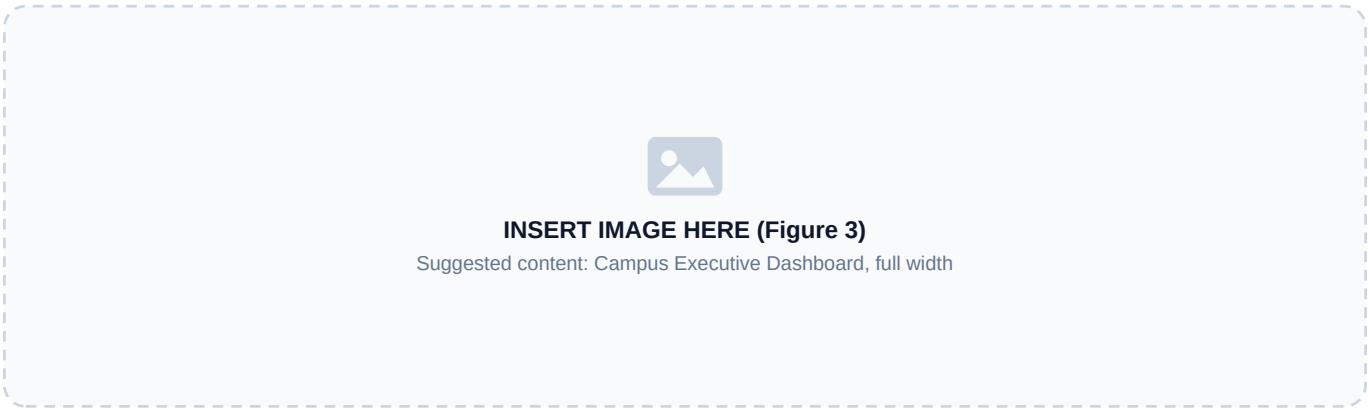


Figure 3: Executive leadership dashboard with KPI widgets, attendance trend and fee donut.

6.1 Widget Reference

Widget	Data source	Visible to
Active students	Students with status Enrolled	Super Admin, Principal
Active faculty and staff	Verified staff directory records	Super Admin, Principal
Today’s attendance	Present plus Late divided by marked students	Super Admin, Principal, Teacher
Total fee collection	Sum of non-cancelled receipts, Decimal(12, 2)	Super Admin, Principal, Cashier
Attendance trend line	Monthly attendance rate for the last 12 months	Super Admin, Principal
Fee donut	Invoice status: Completed, Pending, Overdue	Super Admin, Principal, Cashier
Application feed	Latest enquiries and their statuses	Super Admin, Principal
Live fleet summary (new)	Buses currently on a trip, late or offline	Super Admin, Principal, Transport Coordinator
Career guidance status (new)	Class 12 assessments started, completed and approved	Super Admin, Principal, Career Counsellor

6.2 Quick Actions

- **Admissions & Roster:** enroll new students, link guardians and review pending enquiries.
- **Attendance Register:** mark section rolls, track leaves and process correction tickets.
- **Fee Cashier & Receipts:** process payments, print sequential receipts and manage student dues.
- **Excel Export Center:** one-click spreadsheet exports for statutory and internal reporting.

7. Module 2: Student Lifecycle, Admissions & KYC Directory

PHASE	PRIMARY ROLES	STATUS
Phase 3	**Super Admin, Principal, Office staff**	**Implemented & Verified**

Manages prospective candidate intake from public web enquiry to formal enrollment, KYC records, and parent/guardian pairing.

- **Public Enquiry Intake:** self-service portal submitting reference codes (ENQ-YYYY-XXXXX) without requiring login.
- **Enquiry-to-Admission Workflow:** single-click conversion with auto-generated student admission numbers (ADM-YYYY-XXXXX).
- **Guardian & Sibling Relationships:** maps multiple children to parents for automatic sibling discount qualifications.
- **KYC Digital Vault:** verifies birth certificates, transfer certificates, and passport photos.

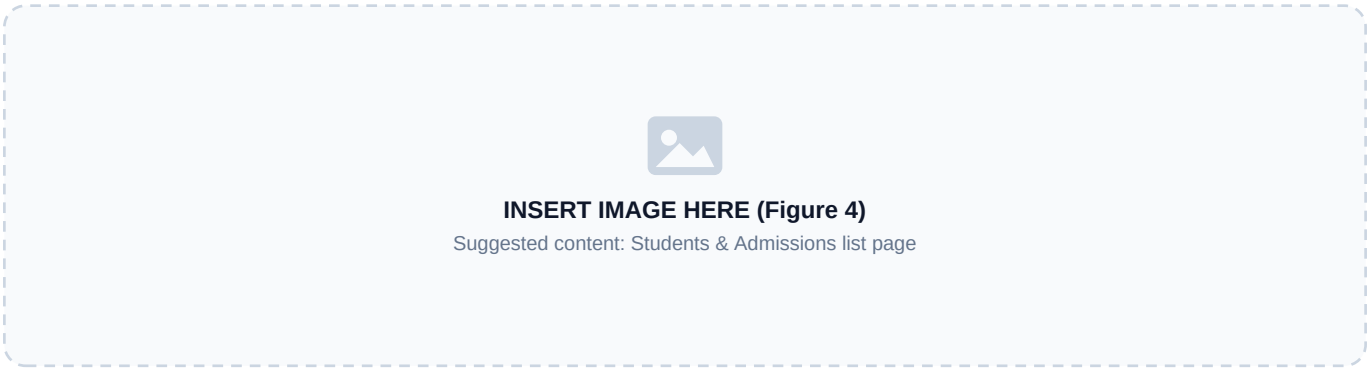


Figure 4: Student directory with grade filters.



Figure 5: Student Admission and KYC modal form.

7.1 Admission Workflow

1. A parent or visitor submits the public enquiry form and receives an enquiry reference (ENQ-YYYY-XXXXX).
2. Office staff review the enquiry and move it through Pending, Interview and Enrolled statuses.
3. On approval, one click converts the enquiry into a student record and generates the admission number (ADM-YYYY-XXXXX).
4. Staff link one or more guardians, upload KYC documents and assign class, section and academic year.
5. The guardian account becomes able to sign in to the Parent Portal for that child only.

7.2 Identifier Formats

Code	Format	Generated when
Enquiry reference	ENQ-YYYY-XXXXX	A public enquiry is submitted
Admission number	ADM-YYYY-XXXXX	An enquiry is converted to a student
Receipt number	RCP-YYYY-XXXXX	A fee payment is recorded (see Module 5)

7.3 KYC Document Checklist

Document	Purpose	Validation rule
Birth certificate	Verify date of birth and parentage	Mandatory; file type and size checked
Transfer certificate	Confirm previous school and class	Mandatory for transfer admissions
Passport photo	Student identity on records and report card	Mandatory; image format checked

7.4 Business Rules

- A guardian may be linked to several children; each child keeps a separate record, class and fee ledger.
- Sibling relationships drive the automatic sibling concession applied by the fee module.
- Student status changes (enrolled, transferred, withdrawn) are recorded with the user and time.
- Students in Class 12 are automatically eligible for Module 10 once their marks are published.

8. Module 3: Classroom Attendance Register & Conflict-Free Timetable

PHASE	PRIMARY ROLES	STATUS
Phase 4	**Teacher, Principal, Super Admin**	**Implemented & Verified**

Eliminates paper roll-calls and scheduling conflicts using automated conflict detection and daily attendance lockdowns.

- **Interactive Roll-Call Grid:** one-click attendance recording with Present, Absent, Late, and Excused tags.
- **Attendance Lockdown & Audit:** automatic evening lock; modifications require principal-signed audit tickets.
- **Conflict-Free Matrix Engine:** algorithmic validation that mathematically rejects teacher or room double-bookings.
- **Real-Time Class Rate:** instant telemetry displaying percentage presence per class section.



Figure 6: Daily attendance roll-call register.

8.1 Attendance Statuses

Status	Meaning	Counted as present?
Present	Student attended the session	Yes
Late	Student arrived after the start time	Yes, flagged as late
Excused	Absence approved with a valid reason	No, but not counted as unexcused
Absent	Student did not attend	No

8.2 Lockdown and Correction Flow

1. The teacher opens the section register for today and marks each student; the class presence rate updates instantly.
2. At the configured evening time the register locks and no further edits are possible from the teacher account.
3. To change a locked record, the teacher raises a correction ticket with a reason.
4. The Principal reviews and signs the ticket; only then is the change applied and written to the audit log with old and new values.

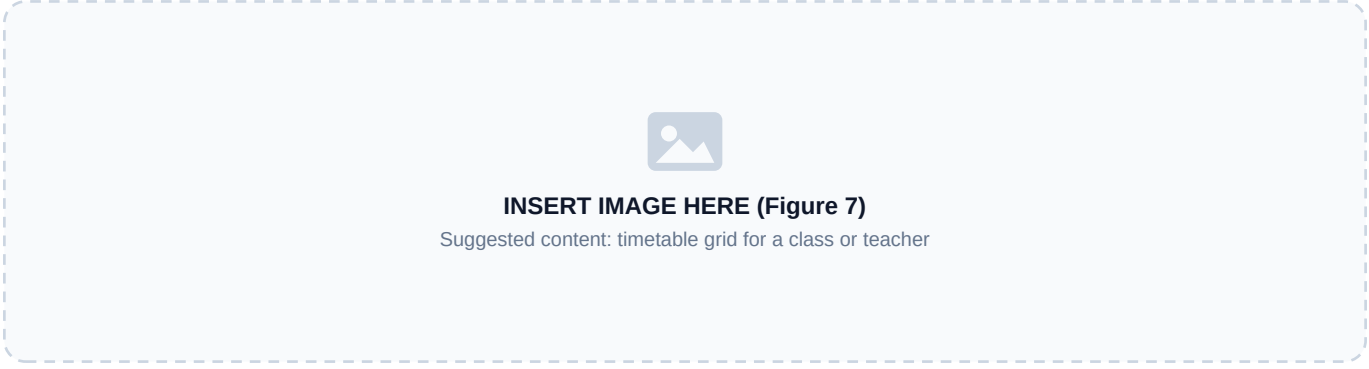


Figure 7: Weekly conflict-free timetable matrix.

8.3 Timetable Conflict Rules

- **Teacher conflict:** a teacher cannot be assigned to two sections in the same day and period.
- **Room conflict:** a room or laboratory cannot host two sessions in the same slot.
- **Section conflict:** a section cannot have two subjects in the same period.
- **Immediate feedback:** the matrix highlights the clashing cell and explains the reason before saving.

9. Module 4: Examinations, Grading Pipeline & PDF Report Cards

PHASE	PRIMARY ROLES	STATUS
Phase 5	**Teacher, HOD, Principal, Super Admin**	**Implemented & Verified**

Automates academic assessments, grade calculation, principal review, and official branded report card synthesis.

- **Multi-Tier Result Pipeline:** Draft → Submitted (Teacher) → Reviewed (HOD) → Approved (Principal) → Published.
- **Gradebook Security:** unapproved marks remain hidden from parents/students via API-level authorization.
- **Standardized Grade Scales:** automatic computation of letter grades (A+, A, B, C, F), percentages, and class ranks.
- **ReportLab PDF Report Card:** high-resolution printable PDF with school crest, grades table, teacher remarks, and verification QR code.

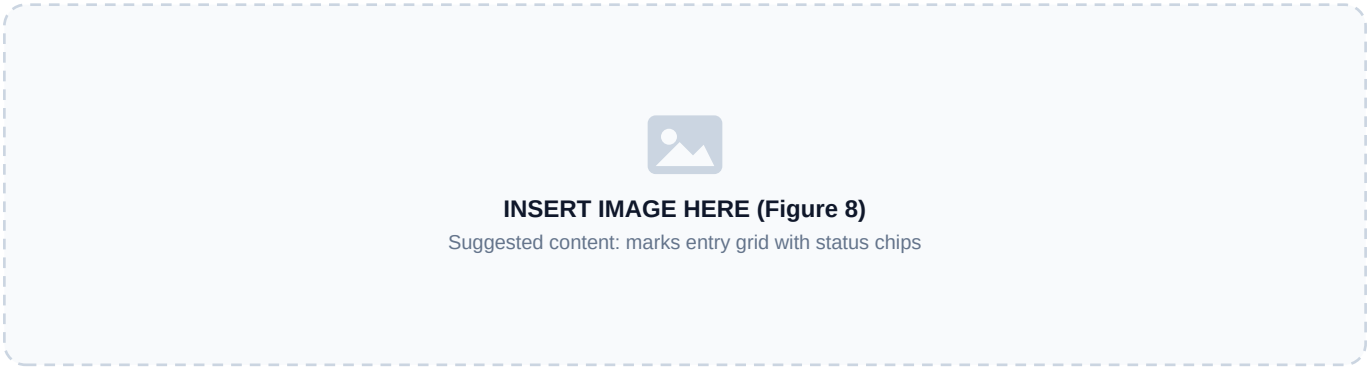


Figure 8: Examination gradebook management screen.

9.1 Result Pipeline

Stage	Actor	Action	Visible to families?
Draft	Teacher	Enters marks for assigned subject and section	No
Submitted	Teacher	Locks entries and submits for review	No
Reviewed	HOD	Checks totals, outliers and remarks	No
Approved	Principal	Gives final approval for the exam	No
Published	Principal / Super Admin	Releases results to portals and enables report card download	Yes, own child only

9.2 Grade Scale (Default, Configurable)

Grade	Percentage band	Remark
A+	90 and above	Outstanding
A	75 to under 90	Excellent
B	60 to under 75	Good
C	40 to under 60	Satisfactory
F	Below 40	Needs improvement

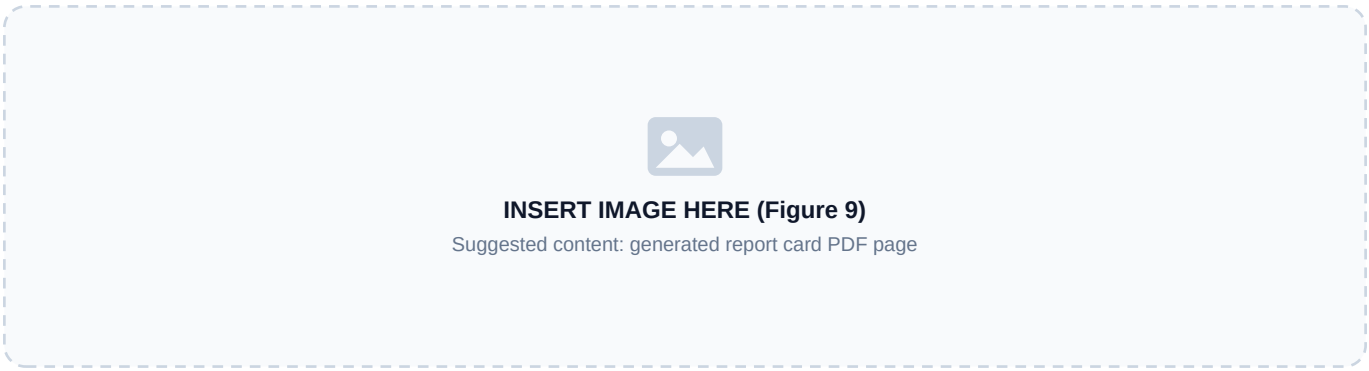


Figure 9: Official ReportLab PDF report card preview.

9.3 Report Card Contents

- School crest, student name, admission number, class, section and academic year.
- Subject-wise marks, percentage, letter grade and class rank.
- Teacher remarks and, where available, the attendance summary for the term.
- A QR verification code so a printed copy can be checked against the system.

10. Module 5: Fee Architecture, Concessions & Sequential Receipts

PHASE	PRIMARY ROLES	STATUS
Phase 6	**Cashier, Principal, Super Admin**	**Implemented & Verified**

A compliant financial management desk featuring granular fee heads, rule-based concessions, and immutable receipt numbers.

- **Granular Fee Heads:** Tuition, Lab, Sports, Library, and Transportation heads mapped to grade levels.
- **Automated Concessions:** sibling discounts (-10%), staff ward concessions, and merit scholarship deductions.
- **Cashier Desk Terminal:** multi-channel support (Cash, Card, UPI) with partial balance installment tracking.
- **Unmodifiable Sequential Receipts:** sequential numbering (RCP-YYYY-XXXXX) backed by DB row locks and PDF download.

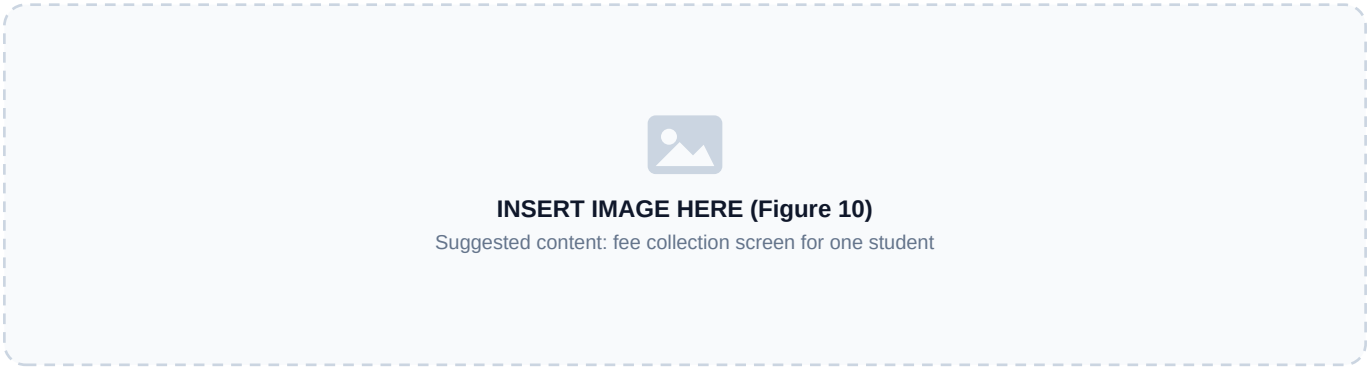


Figure 10: Cashier payment collection terminal with concession breakdown.

10.1 Fee Heads and Concessions

Item	Type	Rule
Tuition	Fee head	Mapped to grade level and academic year
Lab, Sports, Library	Fee head	Applied per grade according to the fee structure
Transportation	Fee head	Applied for students with a bus route (linked to Module 9)
Sibling discount	Concession	-10% automatically applied when siblings are linked
Staff ward concession	Concession	Applied for children of staff members
Merit scholarship	Concession	Deduction approved and recorded against the student

10.2 Payment and Receipt Rules

1. The cashier selects the student and sees outstanding heads after concessions.
2. Payment is recorded by mode (Cash, Card or UPI); partial payments create installment balances.
3. The next receipt number is taken under a database row lock so two cashiers can never receive the same number.
4. A PDF receipt with a QR verification code is generated and can be downloaded or printed.
5. A wrong receipt cannot be edited; it is cancelled with a reason, and the cancellation itself is an immutable record.



Figure 11: Sequential PDF receipt preview.

11. Module 6: Academic Structure & Staff Directory

PHASE	PRIMARY ROLES	STATUS
Phase 2 and Phase 3	**Super Admin, Principal**	**Implemented & Verified**

The academic hierarchy and staff directory are the foundation for attendance, timetable, exams and fees. They define the academic year, which classes and sections exist, which subjects are taught and which teacher handles each combination.

- **Academic year:** one active session (for example 2026–2027) with start and end dates; historical years remain readable.
- **Grades and sections:** each grade has one or more sections with a class teacher and a student capacity.
- **Subjects:** subjects are mapped to grades, with the maximum marks and pass marks used by the exam module.
- **Staff directory:** verified faculty and staff records with role, department, phone number and subject or class assignments.
- **Teacher assignments:** a teacher-subject-section mapping drives roll-call access, mark entry rights and timetable generation.

INSERT IMAGE HERE (Figure 12)

Suggested content: Academic & Classes page with sections and staff list

Figure 12: Academic classes and staff directory.

11.1 Hierarchy

Level	Contains	Used by
Academic year	Terms, holidays, active flag	All modules
Grade	Sections, subjects, fee structure	Admissions, fees, exams
Section	Class teacher, enrolled students	Attendance, timetable
Subject	Maximum and pass marks	Exams, report cards, career scoring
Staff member	Role, department, assignments	Login, timetable, approvals

11.2 Rules

- A student belongs to exactly one section per academic year; promotion creates a new enrollment for the next year.
- Deactivating a staff member immediately revokes sign-in but preserves their historical entries and audit records.
- Subject definitions are versioned per academic year so earlier report cards never change.

12. Module 7: Segregated Parent & Student Portals

PHASE	PRIMARY ROLES	STATUS
Phase 7	**Parent, Student**	**Implemented & Verified**

The Parent Portal and Student Portal are separate experiences on the same backend. Every request is resolved through verified links, so a parent can only ever see the children registered against their phone number, and a student only their own record.

Feature	Parent Portal	Student Portal
Attendance	Strict child-linked view for each linked child	Not shown
Fee status	Dues, paid receipts and downloads	Not shown
Report cards	Published report cards for each linked child	Published term grades
Timetable	Not shown	Personal timetable matrix
Homework notices	Not shown	Notices for own class
Bus tracking (new)	Live position, ETA and alerts for the child's bus	Own bus and route details
Career guidance (new)	Read the approved career report	Take assessment and read approved report

INSERT IMAGE HERE (Figure 13)
Suggested content: parent dashboard

Figure 13: Parent portal showing linked children, attendance and fee status.

INSERT IMAGE HERE (Figure 14)
Suggested content: student dashboard

Figure 14: Student portal with timetable matrix and published grades.

12.1 Zero-Leakage Safeguards

- Every parent query joins through the guardian-student link table; an identifier belonging to another child returns a generic not-found response.
- Unpublished marks and unapproved career reports are filtered out at API level, not just hidden in the interface.
- Automated tests attempt cross-child access with valid parent tokens and must fail for every protected route.
- Download links for report cards, receipts and career reports are generated per request and are tied to the signed-in account.

13. Module 8: Reports & Excel Export Center

PHASE	PRIMARY ROLES	STATUS
Phase 8	**Super Admin, Principal, Cashier, Teacher**	**Implemented & Verified**

The Excel Export Center uses OpenPyXL to produce formatted spreadsheets for statutory submissions and internal review. Each export respects the caller's role, supports filters and records who generated it.

Export	Typical filters	Allowed roles
Student roster	Grade, section, status	Super Admin, Principal
Attendance register	Date range, section, status	Super Admin, Principal, Teacher (own classes)
Exam results	Exam, grade, section	Super Admin, Principal
Fee collection and dues	Date range, fee head, mode	Super Admin, Principal, Cashier
Receipt register	Date range, cancelled or active	Super Admin, Cashier
Transport roster (new)	Route, vehicle, trip date	Super Admin, Transport Coordinator
Career guidance summary (new)	Academic year, status, top field	Super Admin, Career Counsellor

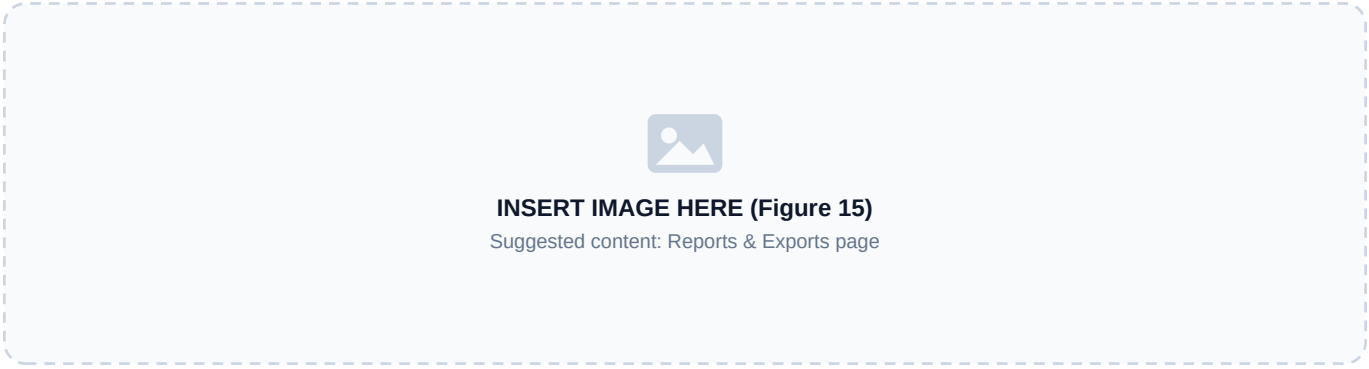


Figure 15: Excel Export Center with filters and download buttons.

13.1 Export Behaviour

- Files are generated on the server with header styling, frozen header row and column widths set for printing.
- Large exports run efficiently with streaming writes so they do not block other requests.
- Every export records the user, filters and time in the audit log.
- Personal data in exports is limited to what the role is allowed to see on screen.

14. Module 9: GPS Bus Tracking & Transport Management (NEW)

PHASE	PRIMARY ROLES	STATUS
Phase 11	**Transport Coordinator, Driver, Parent, Principal**	**Planned**

This module lets the school manage its vehicle fleet and lets parents follow the bus carrying their child. Location can come from a GPS device installed in the bus or from the driver's smartphone. The school sees every vehicle on one live map, and each parent sees only the one vehicle assigned to their child.

14.1 Objectives

- **Safety:** know where every bus is and be alerted early when a trip is delayed, off-route or stopped for too long.
- **Transparency:** parents see the bus position and estimated arrival, reducing waiting time and phone calls.
- **Efficiency:** plan routes and stops once, assign students to them and reuse them every day.
- **Accountability:** trip history records start and end times, stops visited and delays for later review.

14.2 Functional Requirements

ID	Requirement	Priority
GPS-01	Maintain vehicle records: registration number, capacity, make, insurance and fitness expiry dates, assigned GPS device.	Must
GPS-02	Maintain driver and attendant records with licence number, expiry and contact number.	Must
GPS-03	Define routes with ordered stops, pick-up and drop times and map coordinates for each stop.	Must
GPS-04	Assign students to a route and stop; one student has at most one active route per academic year.	Must
GPS-05	Driver can start and end a trip (morning pick-up or evening drop) from a mobile-friendly view.	Must
GPS-06	Receive location updates from the driver app or a GPS provider and store them with a timestamp.	Must
GPS-07	Show all active vehicles on a live map to authorised staff, with status colours.	Must
GPS-08	Parents see only the vehicle and route assigned to their linked child.	Must
GPS-09	Calculate and display the estimated time of arrival for the next stops.	Should
GPS-10	Raise alerts for approaching stop, arrival at school, late start, route deviation and long halt.	Should
GPS-11	Show a clear Stale or Offline label when no recent location is available.	Must
GPS-12	Keep trip history and playback for authorised staff for the configured retention period.	Should
GPS-13	Link the Transportation fee head to route assignment so fees follow transport usage.	Could

INSERT IMAGE HERE (Figure 16)

Suggested content: map with several bus markers and status list

Figure 16: Live fleet map for the transport coordinator.

14.3 Master Data

Entity	Key fields	Notes
Vehicle	Registration number, capacity, device ID, insurance and fitness expiry	Expiry reminders shown on dashboard
Driver / attendant	Name, phone, licence number and expiry	Driver account is created with the Driver role
Route	Name, direction (pick-up or drop), start and end time, vehicle, driver	Reusable for each school day
Stop	Name, latitude, longitude, order, planned time	Ordered inside a route
Student assignment	Student, route, stop, academic year	Drives parent visibility and fee head

14.4 Location Source Options

Aspect	Option A: Driver smartphone	Option B: GPS hardware device
Cost	No hardware cost; needs a phone and mobile data	Device, SIM and subscription per bus
Accuracy	Good in open areas; depends on phone settings	Generally stable; built for vehicles
Reliability	Driver must keep the app open and the phone charged	Always on while the vehicle is powered
Integration	Browser or installable web app sends coordinates every 10–15 seconds	Provider API or webhook delivers positions to the backend
Recommended use	Pilot phase or small fleet	Full rollout across all buses

Design decision. Both options write to the same location table through one ingestion endpoint, so the school can start with Option A and add Option B later without changing the map, alerts or parent screens.

INSERT IMAGE HERE (Figure 17)

Suggested content: phone-size screenshot of the driver screen

Figure 17: Driver mobile view: start trip, next stop and location sharing status.

14.5 Trip Lifecycle

State	Triggered by	System behaviour
Scheduled	Daily route plan	Trip appears for the driver at the planned time
Started	Driver taps Start Trip	Location sharing begins; parents can see the bus
At stop	Geofence entry or driver confirmation	Stop marked as visited with time; next ETA calculated
Completed	Arrival at final stop or driver taps End Trip	Location sharing stops; trip summary saved
Cancelled / Missed	Coordinator action or no start by deadline	Parents on the route are notified in the portal

14.6 Live Tracking and ETA

- **Update interval:** positions are sent every 10–15 seconds while a trip is active and are batched if the network drops briefly.
- **Map:** vehicle marker, route line, stop markers and the child's own stop highlighted for parents.
- **ETA:** remaining route distance divided by recent average speed, adjusted by planned stop times; shown only when data is fresh.
- **Stale handling:** if no update arrives for 60 seconds (configurable), the marker turns grey and shows the last seen time.
- **Real-time delivery:** open dashboards receive updates through WebSocket or Server-Sent Events with polling as a fallback.

14.7 Geofences and Alerts

Alert	Condition (default)	Recipients
Bus approaching stop	Within 500 m or about 5 minutes of the child's stop	Parents on that stop
Bus arrived at school	Enters the school geofence	Coordinator, Principal
Late start	Trip not started 10 minutes after planned time	Transport Coordinator
Route deviation	More than 300 m away from the route line	Transport Coordinator
Long halt	Stationary for more than 10 minutes outside a stop	Transport Coordinator
Document expiry	Insurance, fitness or licence expires within 30 days	Transport Coordinator

INSERT IMAGE HERE (Figure 18)

Suggested content: parent portal bus tracking screen

Figure 18: Parent view: bus position, ETA and the child's stop.

14.8 Privacy and Access Control

- **Purpose limitation:** location is collected only during active trips and used only for transport operations and safety.
- **Least visibility:** a parent query returns only the vehicle, route and stop linked to their child; other buses are never exposed.
- **Driver transparency:** drivers are told when sharing is active and it stops automatically when the trip ends.
- **Retention:** detailed location history is kept for a configurable period (default 30 days) and then deleted or reduced to trip summaries.
- **Audit:** viewing trip history and exporting location data are recorded in the audit log.
- **Compliance:** data about children is handled with the school's privacy notice and applicable Indian data-protection requirements.

14.9 Data Model (Indicative)

Table	Purpose	Important columns
vehicles	Fleet master	id, registration_no, capacity, device_id, insurance_expiry, fitness_expiry
drivers	Driver and attendant master	id, user_id, name, phone, licence_no, licence_expiry
routes	Reusable route plans	id, name, direction, vehicle_id, driver_id, planned_start
route_stops	Ordered stops	id, route_id, name, lat, lng, stop_order, planned_time
student_routes	Student to route and stop	id, student_id, route_id, stop_id, academic_year_id

Table	Purpose	Important columns
trips	One run of a route	id, route_id, date, state, started_at, ended_at
vehicle_locations	Location pings	id, trip_id, lat, lng, speed, heading, recorded_at, received_at
transport_alerts	Generated alerts	id, trip_id, type, message, created_at, acknowledged_by

14.10 API Endpoints (Indicative)

Method and path	Purpose	Roles
GET/POST /transport/vehicles	List and manage vehicles	TC, SA
GET/POST /transport/routes	Manage routes and stops	TC, SA
POST /transport/assignments	Assign students to route and stop	TC, SA
POST /transport/trips/{id}/start and /end	Start or finish a trip	DR, TC
POST /transport/location	Ingest one or more location pings	DR, device token
GET /transport/live	Current positions of active vehicles	TC, PR, SA
GET /transport/my-child/{student_id}	Position, ETA and stop for a linked child	PA, ST
GET /transport/trips/{id}/history	Trip playback and summary	TC, PR, SA

14.11 Non-Functional Requirements

- Support at least 60 buses sending a position every 10 seconds without affecting other modules.
- Average end-to-end delay from the bus to the parent screen under 15 seconds on a normal mobile connection.
- Location ingestion endpoint authenticated with a driver token or a device key, with rate limiting.
- Map and live view remain usable on a mobile browser; stop and vehicle lists remain usable without the map.

14.12 Acceptance Criteria

1. A coordinator can create a route with stops, assign a vehicle, driver and students, and see the assignment on the parent screen.
2. A driver can start a trip and the vehicle appears on the coordinator map within 15 seconds.
3. A parent can see only the bus linked to their own child; attempts to open another vehicle are rejected.
4. When updates stop for more than 60 seconds the marker becomes stale and no ETA is shown.
5. Approaching-stop and late-start alerts are generated according to the configured rules.
6. Location records older than the retention period are removed by the scheduled cleanup job.

15. Module 10: AI Career Recommendation for Class 12 Students (NEW)

PHASE	PRIMARY ROLES	STATUS
Phase 12	**Student, Career Counsellor, Parent, Principal**	**Planned**

After Class 12 a student must choose between engineering, medicine, science, commerce, law, design, defence, teaching and many other paths. This module combines the student's marks with their interests and preferences to produce a ranked list of suitable career fields, matching courses and entrance-exam guidance. The recommendation is advisory: a counsellor reviews every report before it is released, and the student and parents always make the final decision.

15.1 Objectives

- **Personalised:** recommendations come from the student's own results and interest profile, not from general advice.
- **Explainable:** every recommendation shows why it was suggested, using plain language and visible scores.
- **Actionable:** each field lists courses, entrance exams, skills to build and a simple timeline.
- **Supervised:** a counsellor approves reports and can add notes before students and parents see them.

15.2 Who Can Use It and When

- Class 12 students are the primary users; Class 11 students may take the interest assessment as a preview without a final report.
- The assessment opens after the Principal publishes the relevant term or pre-board marks.
- Parents can read the approved report from the Parent Portal; the counsellor and Principal can see all reports.
- A student may repeat the assessment (default limit: three attempts per academic year); every attempt is stored.

15.3 Input Data

Input	Details	Source
Academic marks	Subject-wise marks of Class 10, Class 11 and Class 12 terms; percentage and trend	Examinations module (published marks only)
Stream and subjects	Science (PCM, PCB, PCMB), Commerce, Arts or Vocational; optional subjects such as Computer Science	Student record
Interest questionnaire	36 statements rated on a five-point scale, grouped into six interest types	Student, about 10 minutes
Skill self-rating	Confidence in maths, language, logic, creativity, communication, practical work	Student
Preferences	Study duration, government or private institution, preferred location, budget range	Student with parent
Co-curricular profile	Sports, arts, clubs, competitions and projects	Student, verified by teacher
Counsellor notes	Qualitative observations and constraints	Career Counsellor

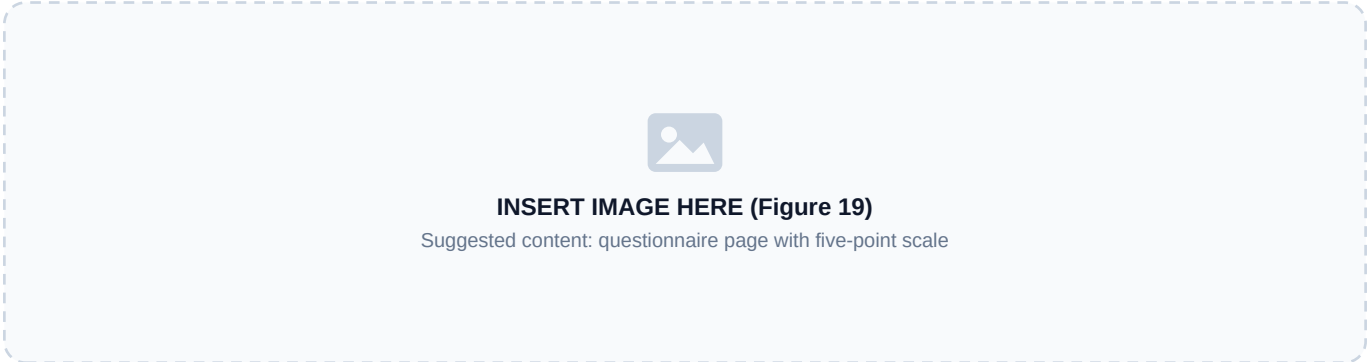


Figure 19: Interest assessment questionnaire for a Class 12 student.

15.4 Interest Assessment (Six Interest Types)

The questionnaire follows the widely used six-type interest model (Holland codes). Each answer adds to one of the six types; the result is a six-number interest profile that is compared with the profile of each career field.

Type	Likes to	Example fields
Realistic	Build, repair and work with tools, machines or outdoors	Engineering, agriculture, aviation
Investigative	Analyse, experiment and solve complex problems	Medicine, research, data science
Artistic	Create, design and express ideas	Design, media, architecture, literature
Social	Teach, help and care for people	Teaching, psychology, nursing, social work
Enterprising	Lead, persuade and start ventures	Management, law, business, public service
Conventional	Organise data, follow clear procedures	Accounting, banking, administration

15.5 Career Field Catalogue

The catalogue is stored in the database and is maintained by the Super Admin and Career Counsellor. Exam names, eligibility and dates change from year to year, so entries must be reviewed at the start of every academic year.

Career field	Usual subjects	Example programmes	Example entrance or selection routes
Engineering & Technology	Physics, Chemistry, Maths	B.E. / B.Tech	JEE Main/Advanced, state counselling (such as TNEA)
Medical & Allied Health	Physics, Chemistry, Biology	MBBS, BDS, B.Sc Nursing, B.Pharm	NEET-UG and state counselling
Pure & Applied Sciences	PCM or PCB	B.Sc, integrated M.Sc	CUET-UG, IISER Aptitude Test, NEST
Computing & Data / AI	Maths preferred, Computer Science	B.Sc CS, BCA, B.Tech CSE / AI	JEE, CUET-UG, university tests
Commerce & Finance	Commerce or any stream	B.Com, CA, CMA, Company Secretary	CA Foundation, CUET-UG
Management & Entrepreneurship	Any stream	BBA, BMS, integrated management	CUET-UG, IPMAT
Law & Public Policy	Any stream	BA LLB, BBA LLB	CLAT, AILET
Humanities & Social Sciences	Arts or any stream	BA Psychology, Economics, Journalism	CUET-UG, university tests
Design, Arts & Media	Any stream	B.Des, BFA, animation, architecture	UCEED, NID DAT, NIFT entrance, NATA
Education & Teaching	Any stream	Integrated B.Ed, B.El.Ed	NCET, university entrance
Defence & Public Services	Any stream; PCM for some wings	NDA, civil services after graduation	NDA exam, UPSC after degree
Agriculture & Environment	PCB or PCM	B.Sc Agriculture, Food Technology	CUET-UG, state agricultural universities
Hospitality & Tourism	Any stream	B.Sc Hospitality, BHM	NCHM JEE, university tests

15.6 Recommendation Pipeline

1. **Collect:** gather published marks, stream, questionnaire answers and preferences for the student.
2. **Filter:** apply hard eligibility rules such as required subjects; ineligible fields are removed or marked as needing a subject change.
3. **Score:** calculate a match score for every remaining field using the weighted formula below.
4. **Rank:** sort fields by score and keep the top five, plus up to two alternative paths that differ in type.
5. **Explain:** create plain-language reasons from the scores; an LLM may polish wording but never changes ranking or numbers.
6. **Review:** counsellor checks the report, edits notes and approves or returns it.
7. **Publish:** the approved report becomes visible to the student and linked parents and a PDF can be downloaded.

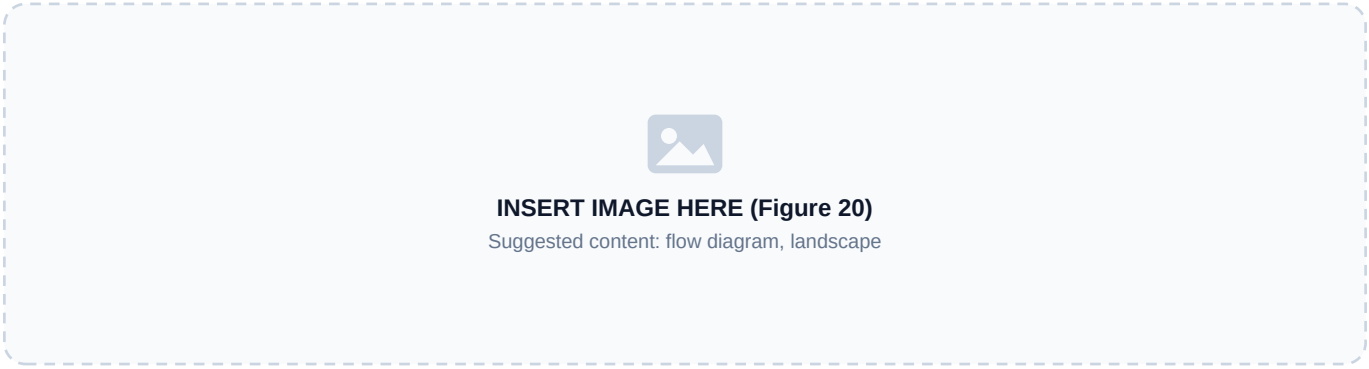


Figure 20: Recommendation pipeline from marks and interests to approved report.

15.7 Scoring Model (Default, Configurable by Super Admin)

Component	Weight	How it is calculated
Academic fit	40%	Weighted average of the subject groups important for the field, compared with the field's typical required level and scaled to 0–100.
Interest fit	35%	Similarity between the student's six-type interest profile and the field's interest profile.
Skill self-rating	10%	Match between self-rated skills and the skills the field needs.
Preference fit	10%	Match with study duration, institution type, location and budget preferences.
Progress trend	5%	Bonus when marks in the relevant subjects are improving across terms.

Match % = (0.40 × Academic) + (0.35 × Interest) + (0.10 × Skill) + (0.10 × Preference) + (0.05 × Trend)

15.8 Illustrative Output

The table below uses sample values only to show the report format; it is not real student data.

Rank	Career field	Match	Reason shown to the student
1	Computing & Data / AI	91%	Strong marks in Maths and Computer Science and a high Investigative interest score.
2	Engineering & Technology	88%	Good Physics and Maths performance with a Realistic and Investigative profile.
3	Pure & Applied Sciences	82%	Consistent science marks and curiosity-driven answers in the questionnaire.
4	Management & Entrepreneurship	74%	Enterprising interest and good communication self-rating.
5	Design, Arts & Media	66%	Creative interest is moderate; marks do not limit this path.

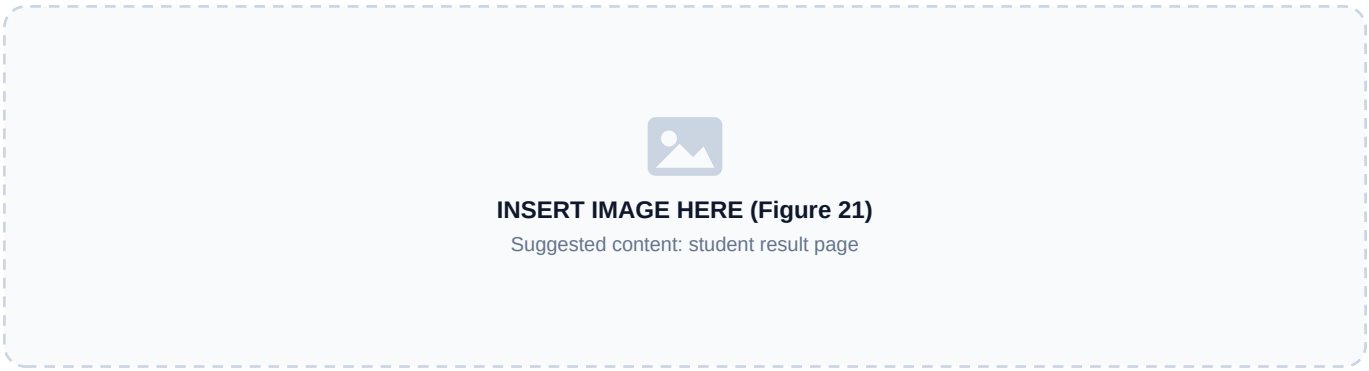


Figure 21: Career recommendation result: top matches with score bars and reasons.

15.9 AI Explanation Layer

- **Grounded:** the model receives only the computed scores, interest profile and catalogue facts, and is told not to invent courses, exams or colleges.
- **Privacy-preserving:** name, phone number, address and admission number are never sent; only anonymised features are used.
- **Validated:** output is checked against a fixed structure; missing or unsafe text triggers a template-based fallback.
- **Replaceable:** the LLM provider is configurable and the module still works with templates if the provider is unavailable.
- **Logged:** the input features, model name, prompt version and output are stored for each report version.

15.10 Career Report (PDF and Portal)

- Student summary: stream, marks overview and subject strengths.
- Interest profile chart with the six types.
- Top five matches with scores, reasons and alternative paths.
- Recommended courses, entrance exams and a month-wise preparation timeline.
- Skills and subjects to improve, counsellor notes, disclaimer and QR verification code.

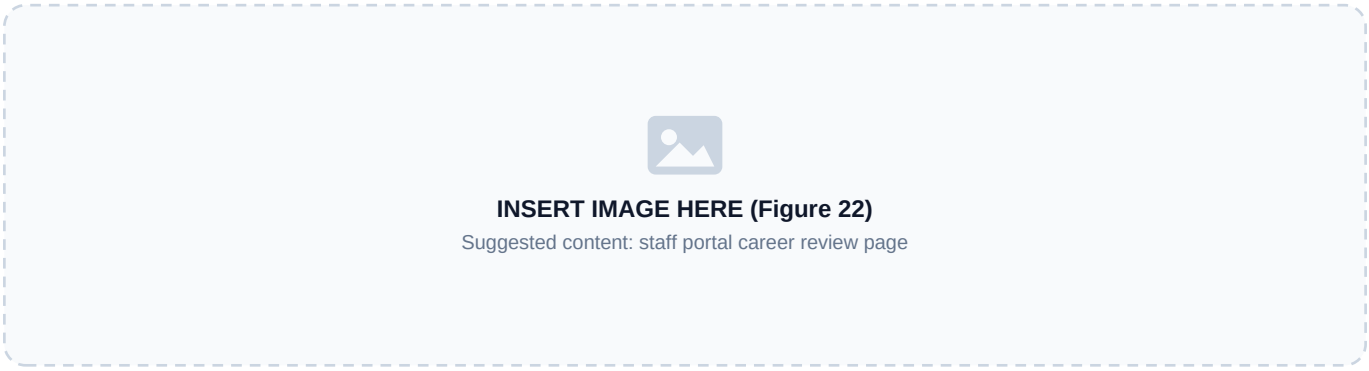


Figure 22: Counsellor review screen with edit, approve and return actions.

15.11 Governance and Responsible Use

- **Human in the loop:** no report is visible to families until the counsellor approves it.
- **Advisory only:** every report states that it is guidance, not a prediction or guarantee of admission or income.
- **Fairness:** gender, religion, caste, community, family income and address are never scoring inputs; budget is used only if the student enters it.
- **Consent:** for students under 18, guardian consent for the assessment is recorded at first use.
- **Data minimisation:** only data needed for guidance is stored; reports are visible to the student, linked parents, counsellor and Principal.
- **Versioning:** each attempt and each regenerated report is stored with its date, weights and catalogue version for audit.

15.12 Data Model (Indicative)

Table	Purpose	Important columns
career_fields	Catalogue of fields	id, name, required_subjects, interest_profile, typical_level, active
career_courses	Programmes and exams per field	id, field_id, course_name, entrance_exam, notes, reviewed_on
assessment_questions	Questionnaire items	id, text, interest_type, order, active
assessment_attempts	One attempt by a student	id, student_id, status, started_at, submitted_at
assessment_answers	Answers to questions	id, attempt_id, question_id, score
career_recommendations	Ranked results	id, attempt_id, field_id, rank, match_pct, component_scores, explanation
counsellor_reviews	Review decisions	id, attempt_id, reviewer_id, status, notes, reviewed_at

15.13 API Endpoints (Indicative)

Method and path	Purpose	Roles
GET /career/questions	Fetch the questionnaire	ST
POST /career/attempts	Start an attempt and submit answers	ST
GET /career/attempts/{id}/report	Read report (only when approved)	ST, PA, CC, PR
GET /career/review-queue	List reports waiting for review	CC, PR
POST /career/attempts/{id}/review	Approve, edit notes or return	CC
GET/PUT /career/fields	Manage catalogue and weights	SA, CC
GET /career/reports/{id}/pdf	Download the PDF report	ST, PA, CC

15.14 Acceptance Criteria

1. A student can complete the questionnaire and see a pending status until approval.
2. The same marks and answers always produce the same ranking and scores (the engine is deterministic).
3. A student whose subjects do not meet a field’s requirements does not receive that field without a clear note.
4. Unapproved reports cannot be fetched by students or parents through the API.
5. No personal identifiers appear in the data sent to the LLM provider.
6. If the LLM provider is unavailable, the report is still produced with template explanations.

16. Database Design

PostgreSQL stores all data in a normalized relational schema. The table names below are logical groupings used to explain the design; the exact table and column names follow the implemented models.

16.1 Design Conventions

- **Primary keys:** UUID v4 on every table through a shared base model.
- **Timestamps:** created and updated times in UTC added by a mixin.
- **Money:** Decimal(12, 2) for every monetary column; no floating-point arithmetic.
- **Constraints:** foreign keys, unique constraints and check constraints enforce integrity in the database itself.
- **Soft history:** financial cancellations and approvals are stored as new records instead of overwriting old ones.

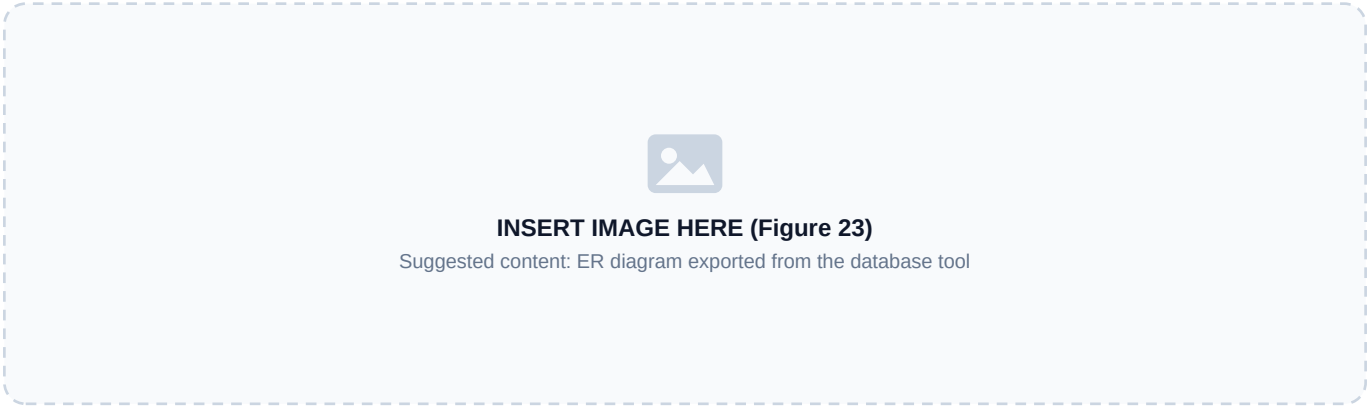


Figure 23: Entity-relationship diagram of the core schema.

16.2 Entity Groups

Group	Key tables (logical)	Notes
Identity and access	users, roles, refresh tokens, login attempts	Lockout counters and token rotation
Academic structure	academic years, grades, sections, subjects, assignments	Foundation for all modules
People	students, guardians, student-guardian links, staff, KYC documents	Links control portal access
Admissions	enquiries, admission records	ENQ and ADM codes
Attendance	attendance sessions, records, correction tickets	Locked daily; changes via tickets
Timetable	periods, rooms, timetable slots	Clash checks before save
Examinations	exams, exam subjects, marks, result states, report cards	Five-stage pipeline
Fees	fee heads, structures, concessions, invoices, payments, receipts, cancellations	RCP numbering with row lock
Transport (new)	vehicles, drivers, routes, stops, student routes, trips, vehicle locations, alerts	See Module 9
Career guidance (new)	career fields, courses, questions, attempts, answers, recommendations, reviews	See Module 10
Audit	audit logs	Who changed what and when

16.3 Integrity and Performance Rules

- Indexed foreign keys on all join columns, with eager loading in the ORM to avoid N+1 queries.
- Receipt numbering uses a transaction with a row lock so numbers are strictly sequential and never reused.
- Location pings are indexed by trip and time, and old rows are removed by a scheduled cleanup job.
- Career recommendations store the weights and catalogue version used, so a past result can be explained later.
- Backups are taken on a schedule and restoration is tested before every major release.

17. API Design

The backend exposes a versioned REST API documented automatically through the FastAPI OpenAPI interface. Each route group maps to a module, uses typed request and response schemas and declares the roles allowed to call it.

- **Versioning:** all routes live under /api/v1.
- **Authentication:** Bearer access token in the Authorization header; refresh endpoint rotates the refresh token.
- **Errors:** consistent JSON error body; authentication failures use one generic message.
- **Pagination and filters:** list endpoints accept page, size and module-specific filters.
- **Idempotency:** payment and location-ingestion calls tolerate retries without duplicating data.

17.1 Route Groups

Group	Base path	Example operations	Main roles
Authentication	/auth	login, refresh, logout, change password	All
Dashboard	/dashboard	KPI summary, attendance trend, fee donut	SA, PR, CA
Students	/students, /enquiries	create, convert, link guardian, upload KYC	SA, PR
Academic	/academic, /staff	years, grades, sections, subjects, staff	SA, PR
Attendance	/attendance	open register, mark, lock, correction ticket	TE, PR
Timetable	/timetable	view, create slot, check conflict	SA, PR, TE, ST
Examinations	/exams	marks entry, submit, review, approve, publish, report card PDF	TE, PR, PA, ST
Fees	/fees	fee structure, invoice, payment, receipt PDF, cancel	CA, SA, PA
Portals	/parent, /student	linked children, published results, dues	PA, ST
Reports	/reports	Excel exports and filters	SA, PR, CA, TE
Transport (new)	/transport	vehicles, routes, trips, location, live view	TC, DR, PA
Career (new)	/career	questions, attempts, review queue, report	ST, CC, PA
Audit	/audit	list and filter audit log entries	SA, PR

17.2 Authorisation Pattern

1. A dependency decodes the token and loads the user and role.
2. A role guard compares the role to the route’s allowed list and returns a generic denial otherwise.
3. For parent and student routes a second guard resolves linked children and rejects any identifier outside the link set.
4. Service functions re-check publication status (results and career reports) before returning data.

18. Non-Functional Requirements & Security

Dimension	Standard	Architectural Implementation
Security & Auth	OWASP Top 10	bcrypt password hashing, JWT Access & Refresh token rotation, account lock after 5 failed attempts.
Performance	Sub-200ms	Indexed foreign keys, eager loading (joinedload) to eliminate N+1 queries, async ASGI pipeline.
Data Integrity	ACID Compliant	High-precision Decimal(12, 2) arithmetic for all finances, strict database constraints and foreign keys.
Traceability	Full Audit Log	Base models with UUID v4 primary keys, UTC timestamp mixins, immutable financial cancellation records.
Deployment	Multi-Platform	Docker Compose for web/cloud services; Electron wrapper with native installer for offline campus PCs.

18.1 Additional Quality Requirements

Area	Requirement
Availability	Web application available during school hours with planned maintenance outside them; target 99% monthly.
Scalability	Supports 3,000 students, 200 staff and about 5,000 parent and student accounts with concurrent peaks at exam result time.
Usability	Responsive layout from phone to desktop; clear labels, consistent navigation and keyboard-friendly forms.
Compatibility	Current versions of Chrome, Edge, Firefox and Safari; Windows desktop through Electron.
Backup and recovery	Daily database backup with tested restore; documented recovery steps.
Maintainability	Typed code, automated tests, module-based structure and written setup documentation.
Localisation	English interface now; text structured so Tamil can be added later.

18.2 Security and Privacy Controls

- **Input validation:** typed schemas on every request; parameterised database access prevents injection.
- **Transport security:** HTTPS in web deployments; restricted CORS origins and secure token handling.
- **Rate limiting:** applied to login, location ingestion and assessment submission.
- **Secrets:** keys and credentials come from environment variables or a secret store, never from source code.
- **Data minimisation:** only necessary personal data is collected; sensitive fields are restricted by role.
- **Child data:** student, location and career data are handled under the school's privacy notice and applicable Indian data-protection requirements.
- **Audit trail:** sensitive reads and writes, exports, approvals and cancellations are logged with user and time.

19. Testing & Quality Assurance

The baseline release is protected by an automated test suite of 34 tests, all passing, together with production builds of the web and desktop applications. The two new modules extend the same approach with additional automated and field tests.

Level	Scope	Approach
Unit tests	Grade calculation, concession rules, receipt sequencing, conflict detection	pytest with isolated test database
API tests	Authentication, RBAC, publication gates, parent-child isolation	Automated requests with tokens for every role
Integration tests	Admission to enrollment, marks pipeline to report card, payment to receipt	End-to-end service flows
Front-end checks	Type checks and production build of the React application	TypeScript compiler and build pipeline
Security tests	Lockout, generic errors, cross-child access attempts	Negative tests and manual review
Field tests (new)	GPS accuracy, signal loss, stale handling on real routes	Pilot with one bus before fleet rollout
Model evaluation (new)	Career scoring stability, eligibility filters, explanation safety	Fixed sample profiles reviewed by counsellors
User acceptance	Daily workflows per role	Pilot with a few teachers, one cashier and volunteer parents

19.1 Test Cases for New Modules

ID	Scenario	Expected result
TC-GPS-1	Driver starts a trip and sends pings every 10 seconds	Bus appears on the coordinator map within 15 seconds
TC-GPS-2	Parent requests another child’s bus	Request is denied with a generic response
TC-GPS-3	Pings stop for 60 seconds	Marker shows Stale and ETA is hidden
TC-GPS-4	Bus passes the approaching-stop radius	One alert is created for parents on that stop (no duplicates)
TC-AI-1	Same marks and answers submitted twice	Identical ranking and scores
TC-AI-2	Student without Biology views Medical field	Field is excluded or flagged with a clear note
TC-AI-3	Student opens report before approval	API returns not available
TC-AI-4	LLM provider disabled	Report is generated with template explanations

19.2 Exit Criteria for a Release

- All automated tests pass and production builds complete without errors.
- No open defects that block a core workflow of any role.
- RBAC and cross-leakage tests pass for every portal.
- Backup restore test is completed and documented.
- Role owners sign the acceptance checklist for their module.

20. Deployment & Infrastructure

The application is delivered in two forms so that the school can use it even when internet access on campus is limited.

Mode	Description	Best for
Web / cloud (Docker Compose)	Backend, database and web front end as containers on a server or cloud VM	Parents, students and remote staff using the portals
Desktop (Electron installer)	Native installer packaging the web app for campus PCs	Cashier and office PCs that must work offline on the campus network

20.1 Environments

- **Development:** local Docker Compose with seeded test data.
- **Staging:** copy of production configuration with anonymised data for UAT and pilots.
- **Production:** hardened configuration, HTTPS, daily backups and monitoring.

20.2 Release Process

1. Merge changes after automated tests pass.
2. Build container images and the Electron installer with a version tag.
3. Deploy to staging and run the smoke and RBAC test checklist.
4. Take a database backup and apply migrations in production.
5. Release to production, watch logs and health checks, and keep the previous version ready for rollback.

20.3 Operations

- **Monitoring:** health endpoint, error logs, slow-query log and disk and memory alerts.
- **Backups:** daily database backups with retention policy and periodic restore drills.
- **Scheduled jobs:** evening attendance lock, location-history cleanup and expiry reminders.
- **Support:** Super Admin manages accounts and audit logs; the development team handles fixes and updates.

21. Implementation & Delivery Verification Status

All ten core developmental phases are completed and verified with automated test suites (34/34 passing) and production builds. Phases 11 and 12 add the new modules described in this version of the document and are planned.

Phase	Module / Description	Verification Status
Phase 1	Project Foundation, Health API & PostgreSQL Setup	**Verified (100%)**
Phase 2	Authentication, RBAC & Academic Hierarchy Models	**Verified (100%)**
Phase 3	Student Admissions, Guardian Linking & Staff Directory	**Verified (100%)**
Phase 4	Attendance Roll-Call & Timetable Clash Detection	**Verified (100%)**
Phase 5	Examinations, Grading Workflow & ReportLab PDF Cards	**Verified (100%)**
Phase 6	Fee Structure, Concessions & Sequential PDF Receipts	**Verified (100%)**
Phase 7	Segregated Parent & Student Portals (No Cross Leakage)	**Verified (100%)**
Phase 8	Role Dashboards & OpenPyXL Excel Statutory Export Center	**Verified (100%)**
Phase 9	Full React 19 + TypeScript + Tailwind Web Architecture	**Verified (100%)**
Phase 10	Desktop Packaging (Electron) & Production Readiness Audit	**Verified (100%)**
Phase 11 (new)	GPS Bus Tracking: master data, trips, live map, alerts, parent view	**Planned**
Phase 12 (new)	AI Career Recommendation: assessment, scoring, report, counsellor review	**Planned**

21.1 Planned Work for Phases 11 and 12

Step	Phase 11: GPS Bus Tracking	Phase 12: AI Career Recommendation
1	Vehicle, driver, route and stop master data	Career catalogue and questionnaire content
2	Student-route assignment and transportation fee link	Attempt, answers and scoring engine
3	Driver trip screen and location ingestion	Eligibility rules and ranking
4	Live map, ETA and stale handling	Explanation layer and PDF report
5	Geofence alerts and parent view	Counsellor review and portal release
6	Field test on one bus, then fleet rollout	Pilot with Class 12 volunteers and counsellor feedback

22. Risks & Mitigations

Risk	Impact	Mitigation
Poor mobile network on bus routes	Missing or delayed location points	Batch and retry pings, show last seen time, mark data as stale instead of guessing
Driver phone is off, low on battery or app closed	Bus appears offline	Driver checklist, power mount, option to move to hardware GPS devices
Privacy concerns about tracking and child data	Loss of parent trust, compliance exposure	Purpose limitation, restricted visibility, retention limits, privacy notice
Career advice seen as a guarantee	Wrong expectations	Advisory wording, counsellor approval, visible reasons and disclaimers
Outdated exam or eligibility information	Misleading recommendations	Yearly catalogue review with owner and date on each entry
Bias or unfair influence in scoring	Unfair recommendations	Exclude sensitive attributes, review weights, test with varied profiles
LLM provider outage or cost growth	No explanations or higher cost	Template fallback, usage limits, replaceable provider
Cross-child data leakage	Serious privacy incident	Linked-record checks, automated negative tests, audit logs
Data loss or server failure	Interruption of daily operations	Daily backups, restore drills, Docker-based redeploy
Low adoption by staff	Parallel paper process continues	Role-based training, quick guides, pilot with champions

23. Future Enhancements

The following items are intentionally outside version 1.1.0 and are recorded for planning. They can be added without redesigning the core platform.

- **Native parent and driver mobile apps:** push notifications and better background location for drivers.
- **Notification channels:** SMS and WhatsApp messages for attendance, dues and transport alerts.
- **Online payments:** payment-gateway integration with automatic reconciliation to receipts.
- **Tamil language interface:** translated labels, report cards and parent messages.
- **Biometric or RFID attendance:** integration for students and staff entry.
- **Payroll, library and inventory:** additional operational modules on the same database.
- **Advanced career insights:** alumni outcome data, scholarship finder and college shortlist tools.
- **Analytics:** predictive alerts for students at risk of low attendance or marks decline.

23.1 Suggested Roadmap

Stage	Focus	Outcome
Stage 1	Phases 11 and 12 delivered and piloted	Live bus tracking and career guidance in use
Stage 2	Notifications and online payments	Fewer phone calls and faster fee collection
Stage 3	Mobile apps and Tamil interface	Wider parent reach and better daily usability
Stage 4	Analytics and optional operational modules	Data-driven planning for management

24. Glossary

Term	Meaning
ERP	Enterprise Resource Planning: one system that connects the school’s daily operations.
PRD	Product Requirements Document: this document.
RBAC	Role-Based Access Control: permissions determined by a user’s role.
JWT	JSON Web Token used to prove identity on each API request.
KYC	Know Your Customer: verified identity and admission documents.
HOD	Head of Department who reviews marks before principal approval.
ASGI	Asynchronous server interface used by FastAPI for concurrent requests.
ORM	Object-Relational Mapper: code layer between Python objects and database tables.
N+1 query	A performance problem where one list query triggers many extra queries.
Geofence	A virtual circle or area on the map that triggers an alert when a vehicle enters or leaves.
ETA	Estimated Time of Arrival of the bus at a stop.
Ping	One location update sent from the driver app or GPS device.
Stale	Location data older than the allowed freshness limit.
Holland codes (RIASEC)	Six interest types: Realistic, Investigative, Artistic, Social, Enterprising, Conventional.
LLM	Large Language Model used only to word explanations, not to rank careers.
Match %	Weighted score between a student’s profile and a career field.
DPDP	India’s Digital Personal Data Protection regime governing handling of personal data.
UAT	User Acceptance Testing: real users confirm that daily workflows work as required.
Idempotent	An operation that gives the same result even if it is repeated, preventing duplicates.
Electron	Framework that packages the web application as a desktop program.
QR code	Square barcode on PDFs used to verify that a printed document is genuine.
Concession	A rule-based reduction in fees such as sibling or scholarship discount.

Appendix A: Figure and Screenshot Checklist

Each placeholder in this document is listed below. Insert the screenshot into the matching dashed box when the final images are ready; use clear, full-width captures and avoid showing real student or parent data.

Figure	Caption	Suggested content
1	High-level system architecture: clients, API, services, database and integrations.	architecture diagram, landscape, about 16:7
2	Unified login portal with role pill tabs and dynamic identifier input.	login screen showing Staff / Parent / Student tabs
3	Executive leadership dashboard with KPI widgets, attendance trend and fee donut.	Campus Executive Dashboard, full width
4	Student directory with grade filters.	Students & Admissions list page
5	Student Admission and KYC modal form.	admission form with guardian and KYC upload
6	Daily attendance roll-call register.	roll-call grid for one section
7	Weekly conflict-free timetable matrix.	timetable grid for a class or teacher
8	Examination gradebook management screen.	marks entry grid with status chips
9	Official ReportLab PDF report card preview.	generated report card PDF page
10	Cashier payment collection terminal with concession breakdown.	fee collection screen for one student
11	Sequential PDF receipt preview.	generated receipt PDF
12	Academic classes and staff directory.	Academic & Classes page with sections and staff list
13	Parent portal showing linked children, attendance and fee status.	parent dashboard
14	Student portal with timetable matrix and published grades.	student dashboard
15	Excel Export Center with filters and download buttons.	Reports & Exports page
16	Live fleet map for the transport coordinator.	map with several bus markers and status list
17	Driver mobile view: start trip, next stop and location sharing status.	phone-size screenshot of the driver screen
18	Parent view: bus position, ETA and the child’s stop.	parent portal bus tracking screen
19	Interest assessment questionnaire for a Class 12 student.	questionnaire page with five-point scale
20	Recommendation pipeline from marks and interests to approved report.	flow diagram, landscape
21	Career recommendation result: top matches with score bars and reasons.	student result page
22	Counsellor review screen with edit, approve and return actions.	staff portal career review page
23	Entity-relationship diagram of the core schema.	ER diagram exported from the database tool

A.1 Image Guidelines

Save each screenshot as figure-01.png, figure-02.png and so on. Use landscape captures of about 1600 by 900 pixels for desktop screens and portrait captures for the driver mobile view. Hide real names, phone numbers and marks before capturing.

Appendix B: Module Traceability Matrix

This matrix links each module to its delivery phase, the data it owns and the checks that prove it works.

Module	Phase	Main data owned	Key verification
0. Unified login	2, 9	users, roles, tokens, login attempts	Lockout after 5 failures; generic errors; token rotation
1. Executive dashboard	8	Read-only aggregates	KPI values match source records
2. Admissions and KYC	3	enquiries, students, guardians, KYC documents	Enquiry converts to ADM number; guardian links created
3. Attendance and timetable	4	attendance records, tickets, timetable slots	Lock at evening; teacher and room clash rejected
4. Exams and report cards	5	exams, marks, result states, report cards	Unapproved marks hidden; PDF with QR generated
5. Fees and receipts	6	fee heads, invoices, payments, receipts	RCP numbers sequential; cancellation immutable
6. Academic structure and staff	2, 3	years, grades, sections, subjects, staff	Assignments drive roll-call and mark-entry rights
7. Parent and student portals	7	Links to existing data	No cross-child access in negative tests
8. Reports and export center	8	Export audit entries	Excel files match on-screen totals
9. GPS bus tracking (new)	11	vehicles, routes, trips, locations, alerts	Live position within 15 s; parent sees own bus only
10. AI career recommendation (new)	12	career fields, attempts, recommendations, reviews	Deterministic ranking; approval gate; no identifiers to LLM

Appendix C: Document Approval

By signing below, reviewers confirm that version 1.1.0 correctly describes the scope and requirements of the School ERP Management System.

Role	Name	Signature	Date
Principal			
Project Lead			
Development Team			